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St. Joseph's College of Arts & Science (Autonomous)

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B. Sc DEGREE EXAMINATION-APRIL 2021

BC405S-INTERMEDIARY METABOLISM-II

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL the questions:

I. Choose the best answer (10X1=10)

1. How many pyvuvate formed in glycolysis?

   a) 1 b) 2 c) 3 d) 4

2. Humans are unable to digest

   a) Starch b) Glucose c) cellulose d) Protein

3. Which of the following compounds is responsible for coordinated regulation of glucose and glycogen metabolism?

   a) NAD^+

   b) Fructose 2,6 bisphosphate

   c) Acetyl-CoA

   d) Fructose 1,6 bisphosphate

4. Fat is a _____ lipid

   a) Simple b) Complex c) Derived d) None

5. The main site of gluconeogenesis is

   a) Muscle b) Kidney c) Liver d) cytosol

6. Which of the following amino acid is a both glucogenenic and ketogenic

   a) Lysine b) Trytophan c) Valine d) Nonem of these

7. Which vitamin is associated wiyh fatty acid synthase complex.

   a) Vit B_{12} b) Biotin c) pantothenie acid d) Niacin

8. Transaminase enzymes present in

   a) Liver b) pancreas c) brain d) Kidney

9. Which of the following product is released in alpha oxidation of fatty acids

   a) CO_2 b) Water c) CO d) N_2

10. A nucleoside is a component of

    a) base b) base and sugar c) base sugar and phosphate d)None

II. Say true or false (5X1=5)

11. Lactate is an end product in an aerobic Glycolysis
12. Urea cycle is take place in cytosol only
13. Poly nucleoside is a basic unit of DNA
14. ATP is known as energy currency
15. CO_2 is utilized in Urea cycle.

III. Answer in one sentence (10X1=10)

16. Define Glycolysis
17. Give one example for essential amino acid.
18. How many ATP is produce per turn of TCA cycle?
19. What is Gluconeogenesis?
20. Mention the purine bases
21. Define co-Enzyme
22. What is Uncouplers?
23. Give one example for phospholipid
24. What are the components in ETC

25. Define Transamination

SECTION-B (4X5=20)

Answer any FOUR of the following:

26. Explain about chemiosmotic theory
27. What are steps involving in purine degradation
28. Mention the steps in Urea cycle
29. Explain the about biosynthesis of fatty acids
30. Compare Glycogenesis with Glycogenolysis
31. Define glyconeogenesis and mention the importance of glyconeogenesis

SECTION-C (3X10=30)

Answer any THREE of the following:

32. Write about the process of Biosynthesis of Cholesterol?
33. Write the steps involving in EMP Pathway
34. Describe the process of Glycogen Metabolism
35. Write the general reactions involved in Amino acid Metabolism
36. Describe the denovo pathway of Purine Biosynthesis with its regulation?

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